

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator exercise 02 exercise 004 EXERCISE 10 CAPSTONE PRO-LOGISTICS SQL File 17* SQL_NM RETAIL mila-nmagl SQL File 20* fraud_sqlnm*

SCHEMAS

Filter objects

- accommodation
- car_rental
- company
- employees_db
- fraud
- fraud_detection**
 - Tables
 - transactions
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
- Views
- Stored Procedures
- Functions
- logistics_project
- new
- nit_edu
- payroll_system
- retail
- retail_db

Administration Schemas

Information:

Schema: fraud_detection

```
1 • USE fraud_detection;
2 • SELECT * FROM transactions LIMIT 10;
3
4 -- query1 Fraud vs Normal Count
5 • SELECT Fraudulent, COUNT(*) as Total
6   FROM transactions
7   GROUP BY Fraudulent;
8
9 -- Query 2 - Fraud Percentage
10 • SELECT ROUND(SUM(Fraudulent) * 100.0 / COUNT(*), 2)
11   AS Fraud_Percentage
12   FROM transactions;
```

Result Grid

	Transaction_ID	User_ID	Transaction_Amount	Transaction_Type	Time_of_Transaction	Device_Used	Location	Previous_Fraudulent_Transactions	Account_Age
▶	T1	4174	1292.76	ATM Withdrawal	16	Tablet	San Francisco	0	119
	T2	4507	1554.58	ATM Withdrawal	13	Mobile	New York	4	79
	T3	1860	2395.02	ATM Withdrawal	12	Mobile	Boston	3	115
	T4	2294	100.1	Bill Payment	15	Desktop	Chicago	4	3
	T5	2130	1490.5	POS Payment	19	Mobile	San Francisco	2	57
	T6	2095	2372.04	ATM Withdrawal	15	Desktop	Boston	3	96
	T7	4772	544.81	Bill Payment	2	Tablet	Boston	3	6
	T8	4092	635.75	ATM Withdrawal	13	Tablet	Boston	2	13
	T9	2638	2318.87	Bank Transfer	12	Mobile	San Francisco	4	110
	T10	3169	3656.17	Bill Payment	3	Mobile	Chicago	4	66

transactions 1 x

Read Only Context

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File Edit View Query Database Server Tools Scripting Help

Navigator exercise 02 exercise 004 EXERCISE 10 CAPSTONE PRO-LOGISTICS SQL File 17* SQL_NM RETAIL mila-nmsql SQL File 20* fraud_sqlnm x

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Schema: fraud_detection

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6 FROM transactions
7 GROUP BY Fraudulent;
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9 -- Query 2 - Fraud Percentage
10 • SELECT ROUND(SUM(Fraudulent) * 100.0 / COUNT(*), 2)
11 AS Fraud_Percentage
12 FROM transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Fraudulent	Total
▶	0	47652
	1	2467

Result 2 x

Read Only

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Navigator exercise 02 exercise 004 EXERCISE 10 CAPSTONE PRO-LOGISTICS SQL File 17* SQL_NM RETAIL mila-nmsql SQL File 20* fraud_sqlnm* SQLAddi

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Schema: **fraud_detection**

```
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6 FROM transactions
7 GROUP BY Fraudulent;
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9 -- Query 2 - Fraud Percentage
10 • SELECT ROUND(SUM(Fraudulent) * 100.0 / COUNT(*), 2)
11 AS Fraud_Percentage
12 FROM transactions;
--
```

Limit to 2000 rows

Result Grid

Fraud_Percentage
4.92

Filter Rows: Exports: Wrap Cell Content:

Result 3 x

Read Only Context

Output

Action Output

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

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 - Triggers
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 - logistics_project
 - new
 - nitt_edu
 - payroll_system
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 - retail_db
- Administration
- Schemas

Information

Schema: **fraud_detection**

exercise 02 exercise 004 EXERCISE 10 CAPSTONE PRO-LOGISTICS SQL File 17* SQL_NM RETAIL mile-nmsql SQL File 20* **fraud_sqlnm***

Limit to 2000 rows

```
7 GROUP BY Fraudulent;
8
9 -- Query 2 - Fraud Percentage
10 • SELECT ROUND(SUM(Fraudulent) * 100.0 / COUNT(*), 2)
11 AS Fraud_Percentage
12 FROM transactions;
13
14 -- Query 3 - Average Amount
15 • SELECT Fraudulent, ROUND(AVG(Transaction_Amount), 2)
16 AS Avg_Amount
17 FROM transactions
18 GROUP BY Fraudulent;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Fraudulent	Avg_Amount
▶	0	2969.31
	1	3097.82

Result 4 x Read Only

Output

Action Output

MySQL Workbench

Local instance MySQL80 x

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Navigator

exercise 02 exercise 004 EXERCISE 10 CAPSTONE PRO-LOGISTICS SQL File 17* SQL_NM RETAIL mila-nmsql SQL File 20* fraud_sqlnm*

Limit to 2000 rows

```
16 AS Avg_Amount
17 FROM transactions
18 GROUP BY Fraudulent;
19
20 -- Query 4 - Fraud by Device
21 SELECT Device_Used, COUNT(*) as Fraud_Count
22 FROM transactions
23 WHERE Fraudulent = 1
24 GROUP BY Device_Used
25 ORDER BY Fraud_Count DESC;
26
27 -- Query 5 - Fraud by Payment Method
```

Result Grid

Device_Used	Fraud_Count
Desktop	887
Mobile	791
Tablet	714
Unknown Device	75

Schema: fraud_detection

Result 5 x

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File Edit View Query Database Server Tools Scripting Help

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Information

Schema: fraud_detection

```
22 FROM transactions
23 WHERE Fraudulent = 1
24 GROUP BY Device_Used
25 ORDER BY Fraud_Count DESC;
26
27 -- Query 5 - Fraud by Payment Method
28 SELECT Payment_Method, COUNT(*) as Fraud_Count
29 FROM transactions
30 WHERE Fraudulent = 1
31 GROUP BY Payment_Method
32 ORDER BY Fraud_Count DESC;
33
```

Result Grid

	Payment_Method	Fraud_Count
▶	UPI	699
	Debit Card	571
	Credit Card	562
	Net Banking	562
	Invalid Method	73

Result 6 x Read Only

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Administration Schemas

Information

Schema: fraud_detection

```
30 WHERE Fraudulent = 1
31 GROUP BY Payment_Method
32 ORDER BY Fraud_Count DESC;
33
34 -- Query 6 - Fraud by Location
35 SELECT Location, COUNT(*) as Fraud_Count
36 FROM transactions
37 WHERE Fraudulent = 1
38 GROUP BY Location
39 ORDER BY Fraud_Count DESC
40 LIMIT 10;
41
```

Result Grid

Location	Fraud_Count
Boston	424
Chicago	329
Los Angeles	305
San Francisco	294
Miami	293
New York	285
Houston	277
Seattle	260

Result 7 x

Output

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Administration Schemas

Information

Schema: fraud_detection

```
38 GROUP BY Location
39 ORDER BY Fraud_Count DESC
40 LIMIT 10;
41
42 -- Query 7 - Top 10 Highest Fraud
43 SELECT Transaction_ID, Transaction_Amount,
44 Location, Device_Used
45 FROM transactions
46 WHERE Fraudulent = 1
47 ORDER BY Transaction_Amount DESC
48 LIMIT 10;
49
```

Result Grid

Transaction_ID	Transaction_Amount	Location	Device_Used
T6012	49997.8	New York	Unknown Device
T31492	49997.8	Boston	Mobile
T36048	49997.8	San Francisco	Mobile
T9592	49997.8	Chicago	Desktop
T6785	49997.8	Chicago	Desktop
T43617	49997.8	Boston	Desktop
T39335	49997.8	Chicago	Mobile
T17061	49997.8	San Francisco	Unknown Device
T13667	49997.8	Boston	Unknown Device
T39332	49997.8	Miami	Mobile

transactions 8 x

Output

Action Output

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Administration Schemas

Information

Schema: fraud_detection

```
45 FROM transactions
46 WHERE Fraudulent = 1
47 ORDER BY Transaction_Amount DESC
48 LIMIT 10;
49
50 -- Query 8 - Max Min Avg Fraud Amount
51 SELECT
52 MAX(Transaction_Amount) as Max_Fraud,
53 MIN(Transaction_Amount) as Min_Fraud,
54 ROUND(AVG(Transaction_Amount), 2) as Avg_Fraud
55 FROM transactions
56 WHERE Fraudulent = 1;
```

Result Grid

	Max_Fraud	Min_Fraud	Avg_Fraud
1	49997.8	14.31	3097.82

Result 9 x

Read Only Context Help

Output

Action Output

#	Time	Action	Message
8	11:17:47	SELECT Location, COUNT(*) as Fraud_Count FROM transactions WHERE Fraudulent = 1 GROUP BY ...	8 row(s) returned
9	11:18:02	SELECT Transaction_ID, Transaction_Amount, Location, Order_ID FROM transactions WHERE F...	10 row(s) returned